Time warpage:
due to high-speed motion or gravitational acceleration, *see* Time dilation
concept deduced by Einstein, 107
as due to rubbery clocks, 397, 400–401, 403
see also Time machines
Topology:
concepts of 463–65, 464b–65b
as foundation for global methods, 465
use of to prove that singularities must occur, 462–63, 465
why introduced into physics by British, 469–77
abortive introduction in Russia, 471–72

Uhuru X-ray telescope, 311f, 312f, 314, 316
Ultraviolet radiation, 25f
Uncertainty principle, 372, 373b-74b, 375
Unification of all physical laws, 525
Universe:
origin of, *see* Big bang
expansion of, 337
structure of, 301, 302f
ultimate fate of, 355–56; *see also* Big crunch
speculations on travel to another, 456–58, 457f, 473–74, 484
steady-state theory of, 460
Uranium, 199, 220–23, 221b-22b, 225, 230
Uranus, 94

Vacuum fluctuations:
concept of, 430–31, 430b-31b, 491–92
as cause of spontaneous emission of radiation, 431b, 432–33
and virtual particles, 439–45, 440f, 441n, 444b
as form of exotic material, 491–92
as destroyers of time machines, 516–21
VIRGO, 392–93
Virtual particles, 439–45, 440f, 441n, 444b

Wave-particle duality:
concept of, 146
history of ideas on, 147b
illustrated by X-rays and radio waves, 322
and the uncertainty principle, 373b
and electron degeneracy, 146
and vacuum fluctuations, 439
Weak energy condition, 508
White dwarf stars:
mystery of, 142–45
observations of, 142
Chandrasekhar’s computations of, 153–55
properties of, 142, 143f, 150, 154f, 167f
maximum mass determines fates of dying stars, 159–60, 161f, 176–78, 177f
maximum mass, 151–52, 154f, 156b-57b, 161f